

- **Enabling an enterprise data lake or data layer:** A data lake is a central repository to store and capture all data, irrespective of its format and source. A data lake can be leveraged as a single and integrated data repository to enable businesses to ask questions which were never thought of in the past. Data from various departmental silos can be integrated in the data lake to reduce the innovation cycle, time-to-market and hence enhance the customer experience and service.
- **Being cost-effective:** Storing and processing Big Data on traditional enterprise hardware could be very expensive. Hadoop was created to be run on commodity hardware. There is no role for any proprietary hardware or software. Various studies suggest that the cost of storing a terabyte of data per year could be 2x to 3x less on Hadoop based systems as compared to non-Hadoop systems. With falling hardware costs, the cost savings could get even better.

With Hadoop, enterprises follow a methodology to implement their Big Data initiatives, which can be summarised as a Big Data implementation framework.

Big Data implementation framework

This comprises certain guidelines for Big Data initiatives, with various components and processes that are listed below.

Data sources: These could be diverse and spread across the organisation or outside it. Data could originate from various sources like online transaction processing (OLTP) systems, operational data sources (ODS), enterprise data warehouses (EDW), application logs, files, data streams, clickstream data, IoT data, REST APIs, machine data, cloud data or social media data. All this data could be either structured, quasi-structured, semi-structured or completely

unstructured. The data can arrive either in batch mode, real-time mode or near-real-time mode.

Acquire: Enterprises can leverage a wide variety of options for acquiring data from different sources. Data from the sources is first loaded to a raw data zone, which stores the data sets in their original form without any modification. Some of the technologies that can be used for data acquisition include:

- **Apache Sqoop:** A service to export and import data from relational databases to Hadoop
- **Apache Flume:** A service which can be used to stream logs to Hadoop
- **Apache Kafka:** A publish-subscribe messaging system used to capture message streams
- **ETL tools:** Some of the commercial ETL tools that enterprises have already invested in and that have good integration capabilities with Hadoop can be leveraged

Process: Data processing includes transforming data from its raw format. After basic data sanitisation and data quality checks are performed, data can be loaded in a staging zone. After the basic checks are done, sensitive and vulnerable data can be redacted so that it can be accessed without revealing any of the PCI (payment card industry), PHI (personal health information) or PII (personally identifiable information) details. Some of the tools and technologies that can be leveraged include:

- **Apache Hive:** A service that provides relational abstraction on Hadoop data to query huge data sets using Hive-QL (a SQL-like language)
- **Apache Spark:** An in-memory processing framework specially developed for handling large scale data and analytics
- **Apache Storm:** A complex event-processing engine developed for processing streaming data in real-time
- **Apache Drill:** A framework developed for supporting data-intensive distributed applications for

interactive analysis of huge data sets

- **MapReduce:** A Hadoop v1 programming model used to process large data sets in parallel by leveraging distributed computing

Curate: Once the data sets are organised, they are analysed in order to extract value and insights from them. Complex business rules can be applied to the data sets to convert them into a form that is more business friendly. This phase could also include applying machine learning models on the data sets for predictive use cases. For scenarios dealing with artificial intelligence, cognitive APIs can be leveraged to extract additional insights from the data. In addition to the tools and technologies used during the process phase, a few additional technologies that can be leveraged include:

- Machine learning models, which have been trained over a period to predict outcomes for the incoming test data
- **Cognitive APIs:** These are sets of pre-built home-grown or third party APIs catering to a specific need and business use case

Publish: After the data is processed and analysed, it is pushed to the publish layer, which comprises various components including the following.

- **Enterprise data warehouse (EDW):** All the extracted insights and analytics can be pushed to the EDW to cater to reporting and dashboarding needs
- **Search engine:** Data can be pushed to search engines like Elastic Search or Solr to address keyword based search use cases
- **Message queue:** Data can be made available to downstream applications by publishing the extracted insights to message queues
- **REST APIs:** RESTful APIs use HTTP requests to manage data

In addition to the implementation framework, enterprises also put in place a comprehensive data architecture framework for Big Data called the Lambda Architecture.